

Software Requirements Specification For Budget Bridge Web App

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Version 1

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1 Introduction

This document describes the architecture and design for the Budget Bridge web application being developed for the Commerce Bank group project in the University of Missouri—Kansas City's (UMKC) Software Engineering Capstone (CS-451R) course. Budget Bridge is a personal budgeting web application that allows users to track their income and expenses, so they can manage and improve their spending habits.

The purpose of this document is to describe the architecture and design of Budget Bridge web application.

2 General Design Constraints

2.1 *Mandated Constraints*

1. Cannot use a CMS (Content Management System) like WordPress

2.2 *User Experience Constraints*

1. The web application is usable by anyone, regardless of previous knowledge
2. The text used in the web application UI is readable and neatly aligned
3. The UI incorporates adequate negative space

2.3 *Frontend (UI) Constraints*

1. The web application must use 1 CSS Framework (e.g. Bootstrap, TailwindCSS, etc.)
2. The web application must use any framework/library (e.g. React, Blazor)

2.4 *Database Constraints*

1. The web application must include a database that stores all user passwords through hashing

2.5 *Application Content Constraints*

1. Application must include these things:
 - a. Login/Register Page
 - b. Home Page / Dashboard
 - c. User Profile & Settings
 - d. At least 2 additional Pages

3 Nonfunctional Requirements

3.1 Security Requirements

1. On the Login/Register page, validate user passwords to make sure they are adequately strong security wise

3.2 User Interface

1. Make sure that the overall UI is easy to navigate by including adequate negative space

3.3 Documentation and Testing

1. The web application will employ mainly manual testing
2. The finish web application will be accompanied with these documentation documents
 - a. Project Charter Document
 - b. Requirement Document (the document you're currently viewing)
 - c. Project Plan Document
 - d. System Architecture
 - e. Test Documentation
 - f. Team Effort Estimation
 - g. User Guide

4 Functional Requirements

4.1 Required Features

4.1.1 Use Case: 1

- Fulfills "Login/Register Page" under 2.5 Application Content Constraints

Description: Register

Actors: User/Customer

Preconditions: User does not have an account

Post Conditions: User is registered with username/password

Basic Flow/Main Success Scenario:

1. User selects "Register"
2. User enters email
3. User enters password with correct requirements
4. User verifies password (enters twice)
5. User selects "Create Account"
6. Login to dashboard

Alternate Flow/Extensions

- 1a. Duplicate email is entered
 - 1. Prompt user to login instead
- 2a. Invalid password
 - 1. Give prompt displaying password requirements
- 3a. Invalid password verify
 - 1. Give prompt telling

4.1.2 Use Case: 2

- Fulfills “Home Page/Dashboard” under 2.5 Application Content Constraints

Description: Dashboard

Actors: User/Customer

Preconditions: Dashboard is given financial data from account/database

Postconditions: Spending information and infographics are displayed

Basic Path/Main Success Scenario:

- 1. Scroll through page to see info
- 2. Apply filters/sorting to view desired data
- 3. User is directed to spending, income, and profile tabs from dashboard
- 4. User has option to log out of their account
- 5. Select bell icon to view alerts
- 6. Select Chat bot icon to ask questions
- 7. Click error icon for list of API errors

Alternate Flow/Extensions: None

4.1.3 Use Case: 3

- Fulfills “At least 2 Additional Pages” under 2.5 Application Content Constraints as the 1st additional page

Description: Spending

Actors: User/Customer

Preconditions: User inputs sending data

Postconditions: Expenses displayed in table

Basic Path/Main Success Scenario:

- 1. Add expenses button – set date, category, amount. Write description.
 - 1a. Add expense or cancel
- 2. View total expenses
- 3. View Spending table
- 4. Sort and filter spending table
- 5. Charts adjust accordingly as user adjusts spending
- 6. Delete item from list of exp

Alternate Flow/Extensions: None

4.1.4 Use Case: 4

- 5 Fulfills “At least 2 Additional Pages” under 2.5 Application Content Constraints as the 2nd additional page

Description: Income

Actors: User/Customer

Preconditions: User inputs spending data

Postconditions: Income displayed in table

Basic Path/Main Success Scenario:

1. Add Income button – set date, category, amount. Write description.
 - 1a. Add income or cancel
2. View total income
3. View Income table
4. Sort and filter Income table
5. Charts adjust accordingly as user adjusts Income
6. Delete item from list of income

Alternate Flow/Extensions: None

4.1.5 Use Case: 5

- Fulfills “User Profile & Settings” under 2.5 Application Content Constraints

Description: Profile

Actors: User/Customer

Preconditions: User registers with email/password and provides personal info

Postconditions: Personal info is stored and accessible in Profile page

Basic Path/Main Success Scenario:

1. View User info (name, username, email)
2. Night/Day mode
3. Edit Profile button to make changes to user info
 - 3a. Save changes or Cancel
4. Change Password button
 - 4a. Enter current password, valid new password, confirm new password
 - 4a. Confirm or Cancel
5. View account settings (Account type, active status, join date)

Alternate Flow/Extensions:

1. Incorrect old password
 - 1a. Prompt user to try again (email reset?)
2. New password is invalid
 - 2a. Prompt user to enter valid password and display password rules
3. New password confirmation doesn't match
 - 3a. Prompt user to double-check password and re-enter